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Aerosol optical properties of the supermicron aerosol particles in a European boreal environment

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Introduction



- Quantify the aerosol optical properties at the SMEAR II site in southern Finland.
- The time series and seasonal variation of the extensive and intensive properties for PM_{10} and PM_1 aerosol particles.
- Special emphasis has been placed on supermicron aerosol particles (i.e. $PM_1 < \text{supermicron aerosol particles} \leq PM_{10}$)
- Absence of optical closure between in situ-derived and AERONET-derived AOD values suggests the contribution of missing coarse mode particles.
- SMEAR II provides a unique setting for observing natural aerosol contributions like pollen and spores, which are significant especially during seasonal peaks.



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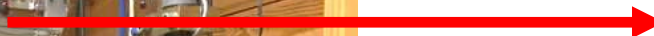
Multi-angle absorption
photometer (Thermo
Scientific MAAP
Model 5012)



Aethalometer (Magee
Scientific model
AE33)



PM₁₀/PM₁ switching
inlet



PM₁₀ range = [(10, 19), (30, 39), (50, 59)]

PM₁ range = [(0, 9), (20, 29), (40, 49)]

Integrating
nephelometer
TSI model 3563



Extensive properties: $\sigma_{\text{abs}, 520}$ and $\sigma_{\text{sca}, 550}$



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Figure 1. Time series of $\sigma_{\text{abs}, 520}$ for PM_{10} , submicron and supermicron aerosol particles from 2010-22.

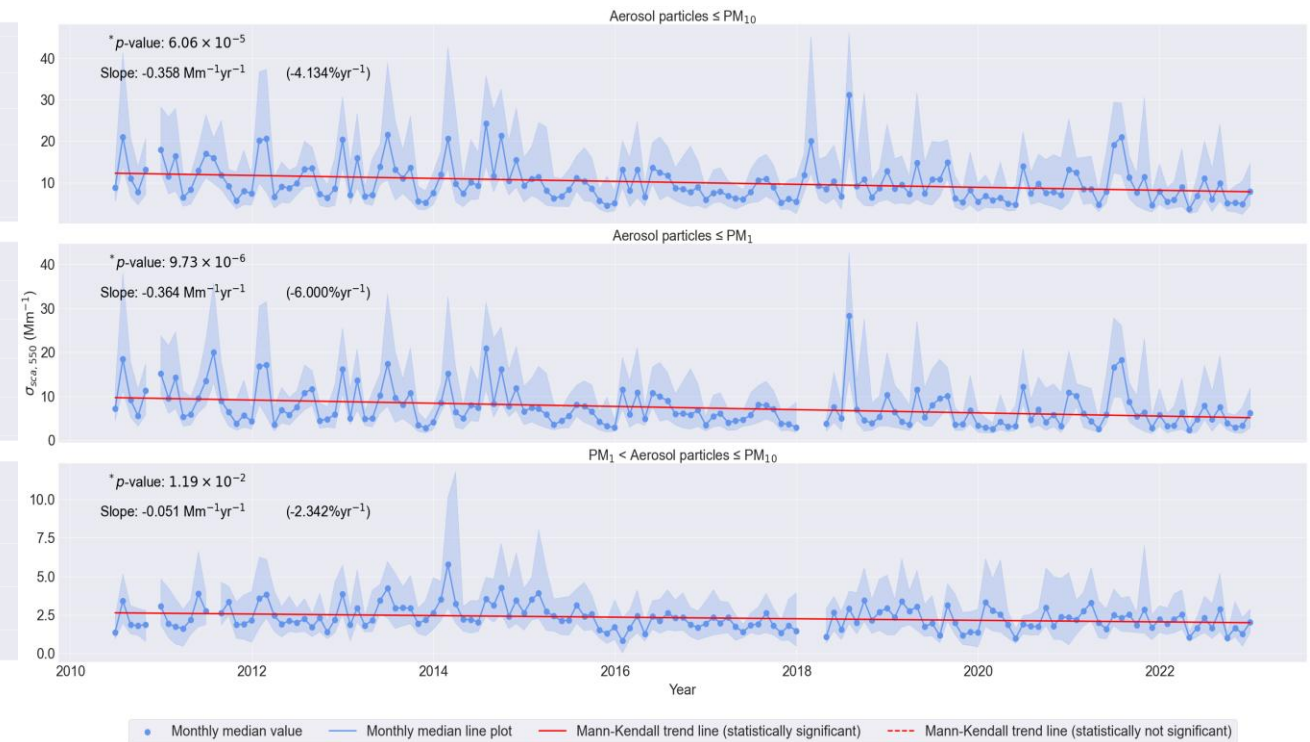
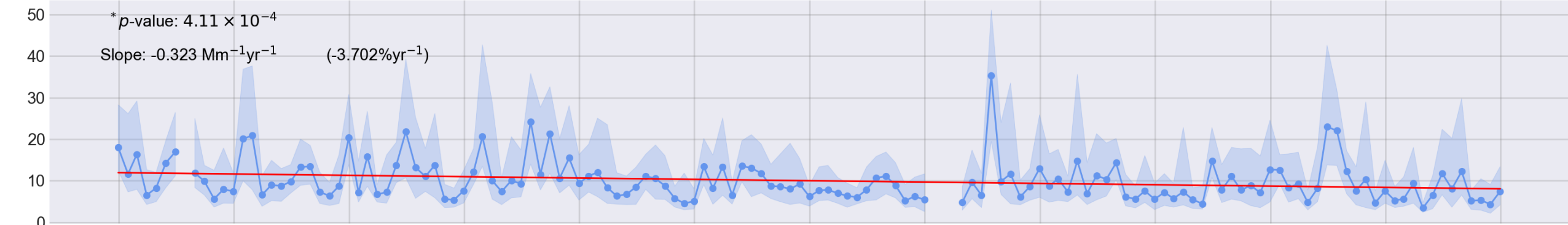
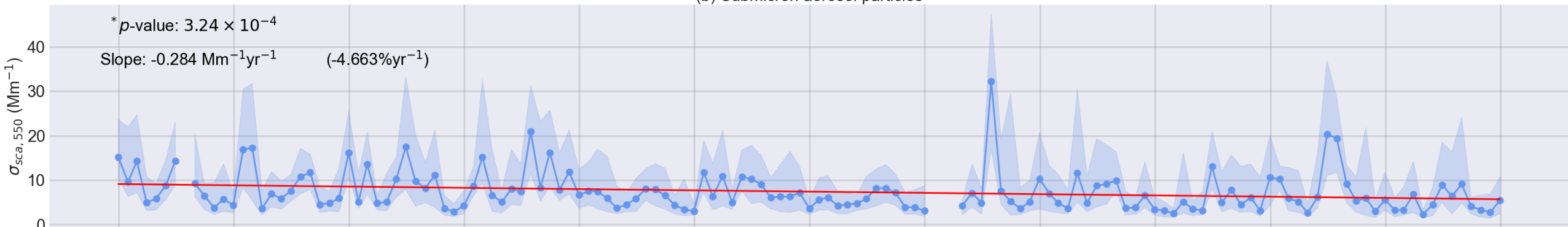


Figure 2. Time series of $\sigma_{\text{sca}, 550}$ for PM_{10} , submicron and supermicron aerosol particles from 2010-22.

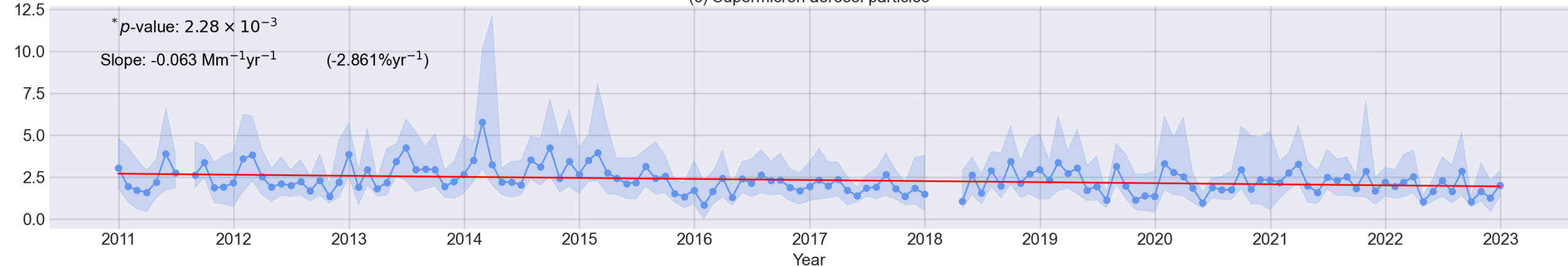
(a) PM₁₀ aerosol particles



(b) Submicron aerosol particles



(c) Supermicron aerosol particles



● Monthly median value — Monthly median line plot — Mann-Kendall trend line (statistically significant) - - - Mann-Kendall trend line (statistically not significant)

Key take away points



- The decreasing trend of the $\sigma_{\text{sca}, 550}$ in the present study, may be partially attributed to a decrease in the concentration of SO_2 since the 1970s (Smith et al., 2011, Klimont et al., 2013).
- Both the reduction of SO_2 between 1998-2018, as reported by Li et al., 2023, and CS between 2006-2022, as reported by Sulo et al., 2022, suggest a decrease in the $\sigma_{\text{sca}, 550}$ of the aerosol particles across the different size ranges.
- The observed decrease in the absorption coefficient was about twice as significant as the decrease in the scattering coefficient.



Supporting indicators from recent literature

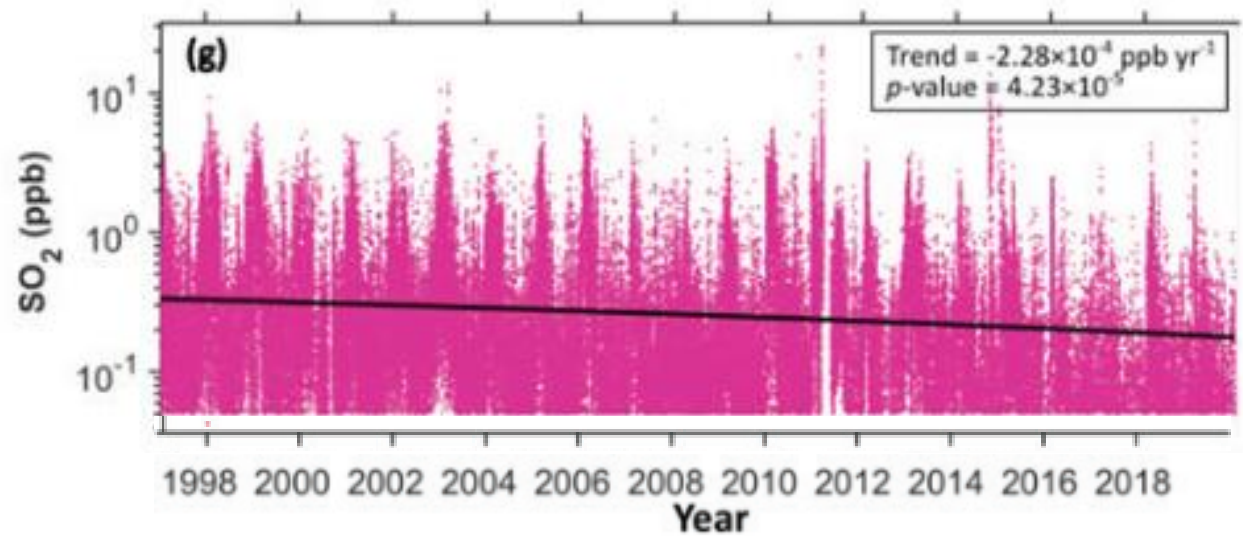


Figure 5. Time series of SO₂ mixing ratio from 1997-2019 at SMEAR II (Li et al., 2023; BER).

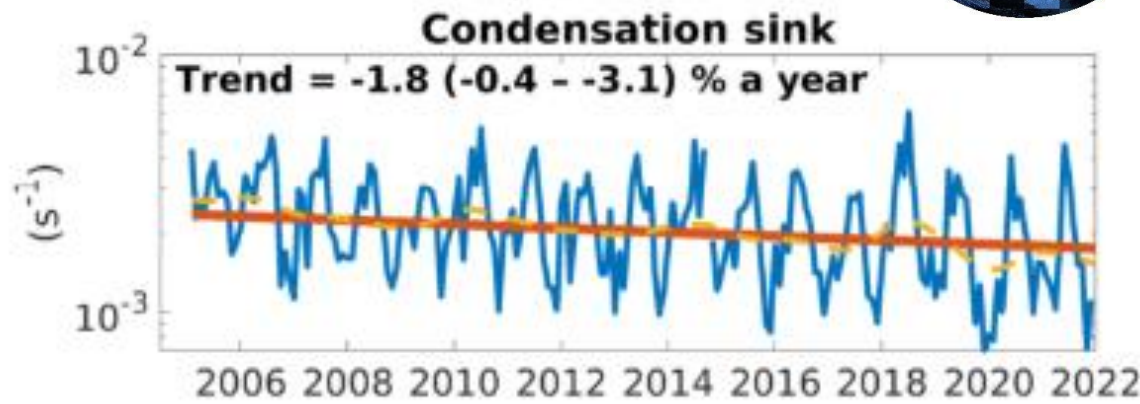


Figure 6. Time series of CS from 2006-2022 at SMEAR II (Sulo et al., 2022; ACP).

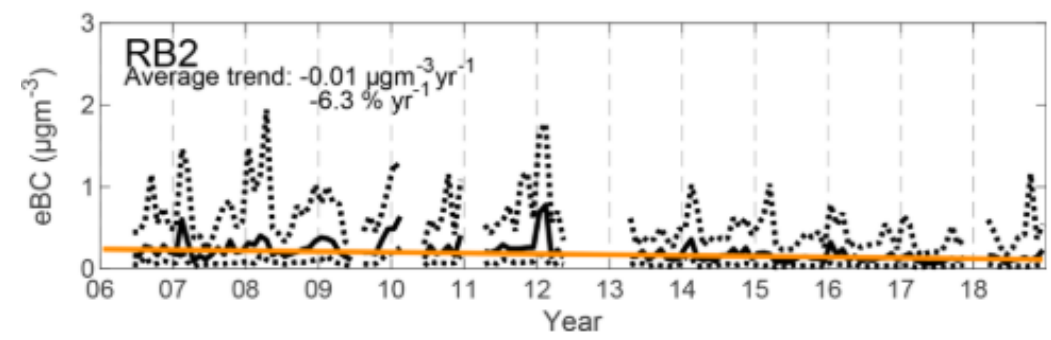


Figure 7. Time series of eBC from 2006-18 at SMEAR II (Luoma et al., 2021; ACP).



Extensive properties: $\sigma_{\text{bsca}, 550}$ and PM mass



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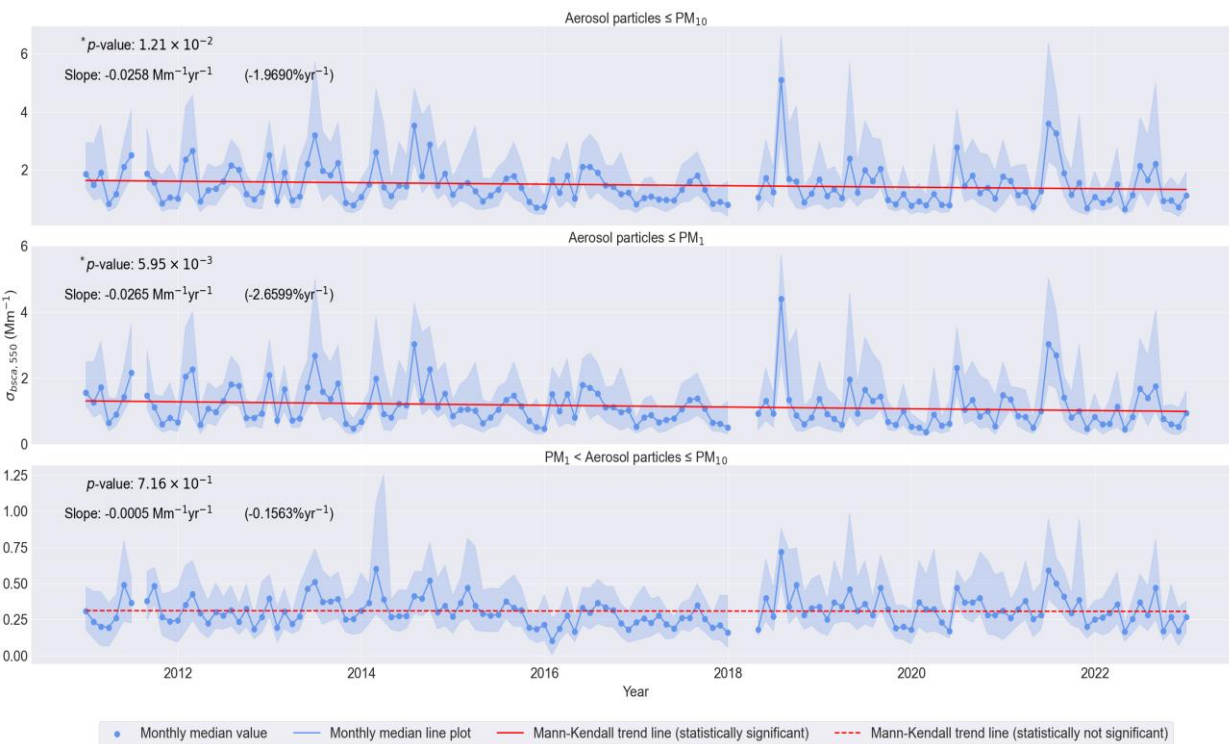


Figure 3. Time series of $\sigma_{\text{bsca}, 550}$ for the PM_{10} , submicron and supermicron aerosol particles from 2010-22.



Figure 4. Time series of PM mass for the submicron, supermicron and super- PM_{10} aerosol particles from 2010-22.



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Intensive properties: AAE and SAE

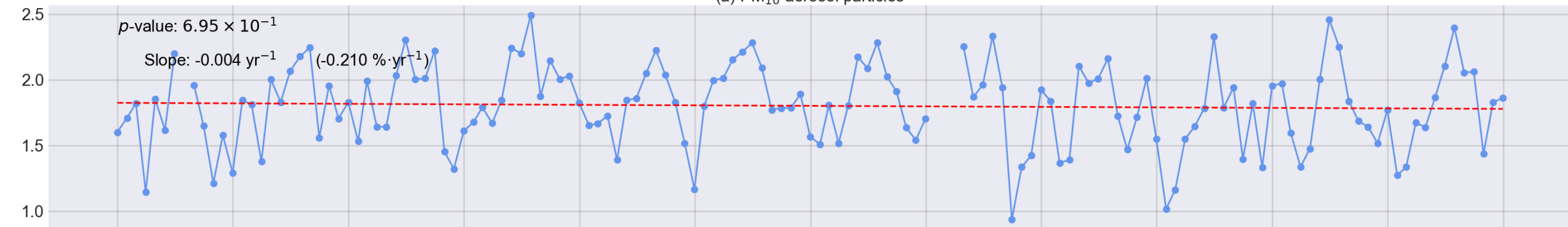


Figure 7. Time series of AAE from 2010-22 at SMEAR II.



Figure 8. Time series of SAE from 2010-22 at SMEAR II.

(a) PM₁₀ aerosol particles



(b) Submicron aerosol particles



(c) Supermicron aerosol particles



2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023

Year

● Monthly median value — Monthly median line plot — Mann-Kendall trend line (statistically significant) - - - Mann-Kendall trend line (statistically not significant)



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Implications for aerosol size distribution



- An increasing SAE of the submicron aerosol particles suggests a shift in the accumulation-mode particles toward smaller sizes.
- This is further emphasized by the increasing trends that is observed in the SSA and the b in the submicron aerosol particles.

Intensive properties: SSA and b

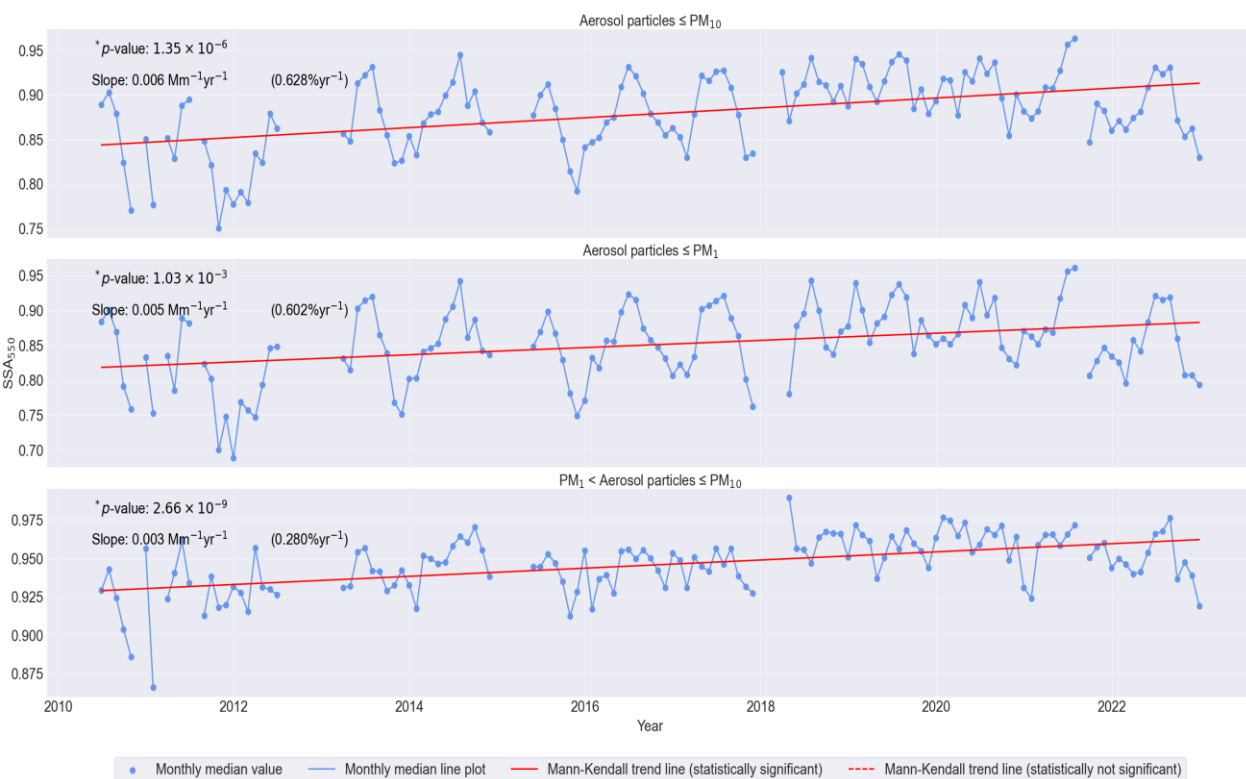


Figure 8. Time series of SSA from 2010-22 at SMEAR II.

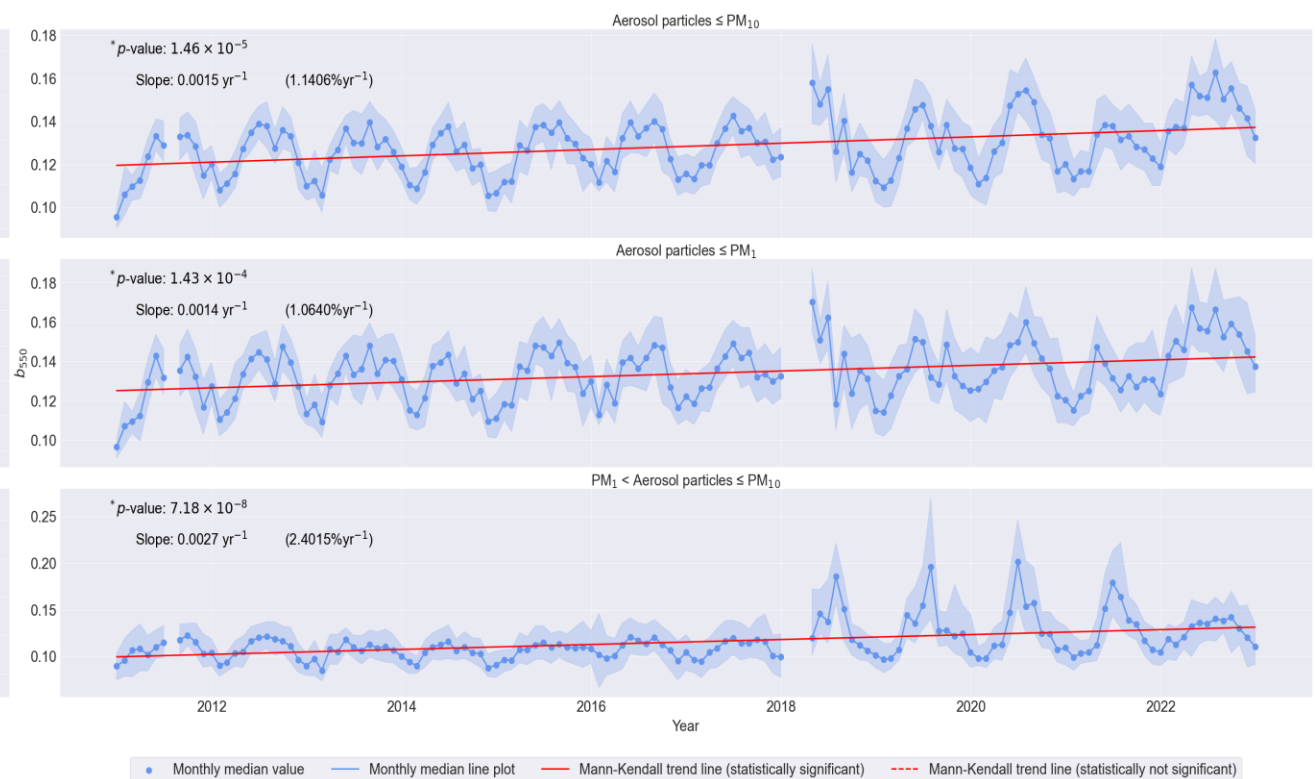


Figure 9. Time series of b from 2010-22 at SMEAR II.



Contribution of the supermicron aerosol particles

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Figure 10. Time series of PM mass for the submicron, supermicron and super- PM_{10} aerosol particles from 2010-22.

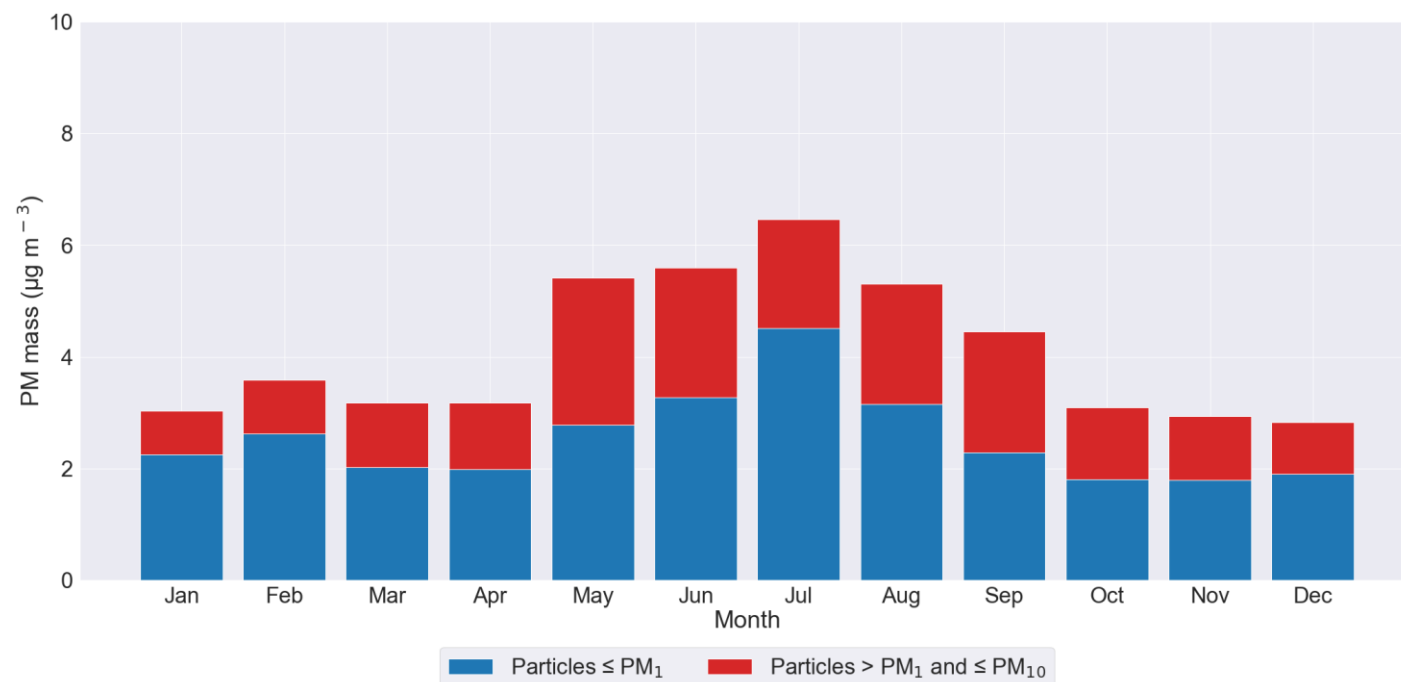


Figure 11. Seasonal variability of PM mass for the submicron and supermicron aerosol particles from 2010-22.



Summary and conclusions



- The overall trend seems to show that the aerosol size distribution consists of more submicron aerosol particles.
- The trend of b for the supermicron aerosol particles is approx. > 2 times the trend of b for the submicron aerosol particles.
- But a very sharp decline in the SAE of the supermicron aerosol particles suggests that perhaps supermicron aerosol particles may have a role to play.
- The seasonal variability of PM mass does show that supermicron aerosol particles can make a substantial fraction of the PM mass during the spring and summer seasons.
- However, optically the supermicron aerosol particles show very little effect.